Social media exposure raises antidepressant demand

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Keywords: social media; mental health; antidepressant expenditure; instrumental variables
Classification: Social Sciences (Social Sciences);
Social Sciences (Psychological and Cognitive Sciences)

Abstract

We provide new evidence on the impact of social media exposure on mental health, by linking 2.5 million antidepressant procurement records and digital traces from nearly 600 social media applications in 323 Chinese cities (2019–2023). A one-hour increase in daily per-capita social media use, 54% of the mean, corresponds to 6% higher expenditure under OLS and 8% under IV, concentrated in first-line medications. Effects are concentrated in user-content-based platforms, such as short-video apps and microblogs, while task-based platforms show no detectable effect. They are also stronger in cities with more vulnerable populations, such as those with younger residents, lower educational attainment, or lower employment or marriage rates. Survey evidence suggests time displacement and social perceptions as potential channels. Population-scale digital exposure predicts clinically meaningful antidepressant demand, with effects concentrated in user-content-based platforms and in socially vulnerable populations.

Significance Statement

We bring clinically grounded, population-level evidence to the study of social media's mental health effects, drawing on the universe of antidepressant procurement records from Chinese medical institutions. We link these records to nationwide usage data covering nearly 600 social media applications and over 1.3 billion mobile devices. Combining these datasets, we provide causal evidence on whether social media use harms mental health. Our findings offer a nuanced view that helps reconcile differences across prior studies, and carry direct implications for platform regulation and mental-health policy.

Introduction

Social media is now one of the most time-intensive activities of modern life: nearly 5 billion people use these platforms, spending more than 2 hours per day on average (*1, 2*). Whether such exposure shapes mental health—and depression in particular, a leading contributor to global disease burden with substantial economic costs (*3, 4*)—is a central question across science and policy. Yet the evidence is mixed. Observational studies are themselves divided: some find negative associations between social media use and mental health (*5, 6*), while others report null or weak links (*7–12*). Quasi-experimental and randomized studies have also linked social media exposure to declines in well-being and mental health (*13–17*). Meta-analyses converge on small or negligible negative associations (*18, 19*), with effects varying across populations (*20*). Whether social media shapes population mental health remains debated.

Three limitations recur across the prior literature and partly explain the mixed evidence. First, mental-health outcomes typically rely on self-reported well-being or symptom indices, which are vulnerable to response-style biases and inattentive responding (*21–24*); such measures capture subjective or subclinical distress rather than medically recognized conditions or realized treatment-seeking. Second, social media exposure is often either self-reported, which introduces additional measurement error (*25*), or measured on a single platform, which misses how usage is distributed across hundreds of applications in a fragmented digital environment. Third, evidence draws from narrow samples—college students, adolescents, or volunteer cohorts observed over short windows—which differ from the general population in baseline mental health and life context and capture transient mood fluctuations rather than sustained changes in clinical demand. Closing these gaps requires evidence that combines all three: clinically grounded outcomes, cross-platform exposure, and population-scale coverage.

We address these gaps with two nationwide datasets from China—administrative antidepressant procurement records and population-level social media usage data—both covering the entire country. The procurement data cover more than 2.5 million transactions from roughly 37,000 medical institutions—the universe of antidepressant procurement nationwide. Procurement serves as a proxy for near-term dispensing because medical institutions have little incentive to stockpile. It captures aggregate demand across the population rather than usage in any specific subgroup. The

social media data, from China's three major telecommunications carriers, passively record real-time activity across nearly 600 applications at city × month × gender × age granularity—cross-platform exposure at population scale rather than activity on a single platform. Together, the two datasets let us test whether changes in population social media exposure translate into changes in antidepressant demand.

China—the largest upper-middle-income country—is where evidence on social media and mental health is thinnest and depression is heaviest: about 40 million adults, roughly 13% of the 300 million worldwide (*3, 26*).

In this setting, a one-hour increase in daily per-capita social media use is associated with a 5.6% increase in per-capita antidepressant expenditure (OLS; city and year-month fixed effects; time-varying controls; 323 cities, January 2019–April 2023). The increase is concentrated in first-line medications and is much smaller for second-line medications, consistent with increased demand for initial treatment rather than drugs reserved for complex, recurrent, or treatment-resistant cases.

We identify causal effects with a shift-share (Bartik) instrument for city-level social media exposure (*27–29*). Under IV, a one-hour increase raises per-capita antidepressant expenditure by 7.8%, again concentrated in first-line medications. A drug-by-drug analysis using dosage as the outcome—which removes any confounding from price or product-mix shifts—confirms the pattern: all eight first-line drugs show positive effects, whereas second-line drugs mostly cluster near zero. Because both exposure and demand are measured at the population level, the estimates capture how average demand differs across populations with different levels of exposure, rather than how any individual responds to their own use over time.

Effects vary along two dimensions: platform type and population vulnerability. On platform type, effects are large for user-content-based apps—short-video, microblog, community and forum, and instant messaging, where users consume, produce, and share user-generated content—and absent for task-based apps such as dating, professional networking, and other social-media apps, which are used for specific social or professional purposes. On population, effects are larger in cities with more vulnerable populations—those with younger age structures, lower educational attainment, weaker labor-market attachment, or lower marriage rates—than in others, where estimates remain positive but smaller. Both patterns may partly explain the mixed findings in prior work.

Nationally representative survey data point to two mechanisms—*time displacement* and *social*

perceptions. Social media use is associated with less time on offline activities that support mental health, and with higher perceived inequality, greater income underestimation, weaker social trust, and lower confidence about the future.

Data

Medication procurement data. We obtain order-level administrative records of antidepressant procurement from China's National Drug Procurement Platform, which captures every purchase by medical institutions in mainland China. We restrict the records to antidepressant medications under the World Health Organization's Anatomical Therapeutic Chemical (ATC) code N06A. The resulting sample covers 17 generic compounds and 2,512,878 transactions by 36,955 medical institutions across 323 prefecture-level cities, from January 2019 through April 2023. To validate this procurement-based measure externally, we compare city–year per-capita antidepressant expenditure with self-reported depressive symptoms from the Chinese General Social Survey (CGSS). Respondents in city–years with higher expenditure report more frequent depressive symptoms ($p < 0.001$; Fig. S1).

We use procurement as a proxy for realized treatment demand. First, antidepressants are prescription-only medications, so institutional procurement reflects physician-prescribed treatment rather than discretionary consumer choice. Second, medical institutions face limited economic incentives to stockpile: inventory carries storage costs, and finite drug shelf lives constrain accumulation. Procurement therefore closely tracks near-term dispensing. Based on their typical clinical use, we classify eight compounds as first-line medications and nine as second-line. First-line medications typically serve as initial treatment for new or mild-to-moderate depressive symptoms; second-line medications are more often reserved for follow-up treatment or for complex, recurrent, or treatment-resistant cases. The classification details are provided in Section S1.1 of the Supplementary Materials and Table S1.

Social media data. We obtain population-level digital traces from a leading Chinese internet analytics provider. In partnership with the country's three major telecommunications carriers, the provider passively records real-time mobile-application activity on more than 83% of mobile devices nationwide. The data cover the full set of nearly 600 social media applications operating in the

Chinese market. For each application, the data report aggregate usage time in each city × month × gender × age cell, covering all 323 cities from January 2019 through April 2023. The age dimension distinguishes nine groups: under 18, 18–24, 25–30, 31–35, 36–40, 41–45, 46–50, 51–60, and above 60. The applications are grouped into seven categories: short-video, microblog, community and forum, instant messaging, dating, professional networking, and other social-media apps. Figure S2 documents the distribution, temporal growth, platform composition, and demographic profile of social media use.

Other data. We also draw on four complementary data sources: the China City Statistical Yearbook for annual socioeconomic controls; China's 2015 1% National Population Sample Survey for the demographic shares used to construct the shift-share instrument introduced below; the nationally representative CGSS for individual-level measures of depressive symptoms, time use, and social perceptions; and city–month Baidu search data for indices of depression awareness, unemployment perceptions, and recession perceptions.

Variable construction. Linking the procurement and digital-trace data by city and month yields a balanced panel of 16,796 city–month observations. The primary outcome is the log of monthly per-capita antidepressant expenditure, calculated for all antidepressants and separately for first-line and second-line medications; compound-level analyses use log per-capita dosage in milligrams for each of the 17 generic compounds. The exposure is daily per-capita social media use in hours, aggregated across all applications and, separately, for two platform types defined below (user-content-based and task-based). Summary statistics appear in Tables S2 and S3; further details on data and variable construction are in Sections S1.1–S1.6 of the Supplementary Materials.

Empirical strategy

Baseline specification. We estimate the effect of social media use on antidepressant expenditure with

$$\ln(Y_{ct}) = \beta\,\mathrm{SMUsage}_{ct} + X'_{ct}\gamma + \mu_c + \lambda_t + \varepsilon_{ct}, \tag{1}$$

where Y_{ct} is per-capita antidepressant expenditure in RMB (total, first-line, or second-line) in city c and year-month t; $\mathrm{SMUsage}_{ct}$ is daily per-capita social media usage in hours; X_{ct} is a vector of time-varying city-level controls drawn from the China City Statistical Yearbook—GDP per capita,

fiscal expenditure per capita, secondary and tertiary industry shares, and hospital beds and doctors per 1,000 residents; and μ_c and λ_t are city and year-month fixed effects. The coefficient β measures the percentage change in per-capita antidepressant expenditure per one-hour increase in daily social media use. Standard errors are clustered at the city level.

Bartik instrument. OLS estimates of β may be biased by unobserved local shocks that drive both social media use and antidepressant demand, or by reverse causality if worse mental health raises social media use. We address these concerns with a shift-share (Bartik) instrument that combines predetermined city demographic composition with leave-one-out use among the same demographic groups in other cities of the same province (*27–29*). Social media diffuses unevenly across demographic groups. Platforms target different audiences through content and promotion, and the propensity to adopt social media differs by age and gender. Cities with larger predetermined shares of fast-growing user groups therefore see greater predicted increases in exposure.

Let p denote the province containing city c, and let $d = (g, a)$ index gender × age cells, where age falls into nine groups from under 18 to above 60. The share s_{cd}^{2015} is city c's 2015 population share in cell d; the shift $\overline{\text{SMUsage}}_{p,-c,t,d}$ is the average daily use of cell d in year-month t across other cities of province p, excluding c. The instrument (i.e., predicted social media use) is defined as follows:

$$Z_{ct} = \sum_d s_{cd}^{2015} \, \overline{\text{SMUsage}}_{p,-c,t,d}. \tag{2}$$

We estimate β by two-stage least squares. The first stage is

$$\text{SMUsage}_{ct} = \pi Z_{ct} + X_{ct}'\rho + \mu_c + \lambda_t + \nu_{ct}, \tag{3}$$

and the second stage is

$$\ln(Y_{ct}) = \beta \, \widehat{\text{SMUsage}}_{ct} + X_{ct}'\gamma + \mu_c + \lambda_t + \varepsilon_{ct}. \tag{4}$$

Both stages include the same controls X_{ct}, city fixed effects μ_c, and year-month fixed effects λ_t; standard errors are clustered at the city level. The first-stage Kleibergen–Paap Wald F-statistic exceeds 500 (Table S4), well above conventional thresholds for weak instruments.

Identifying assumptions. Identification requires that Z_{ct} affects $\ln(Y_{ct})$ only through SMUsage_{ct}. We adopt the shares-based interpretation of shift-share designs (*28*), under which identification rests on the exogeneity of the *shares*—each city's 2015 gender × age composition. This composition

is plausibly orthogonal to antidepressant-demand trends because it is predetermined and largely reflects long-run demographic forces—historical fertility, birth-cohort size, and past migration—rather than sample-period social media use or antidepressant demand. The exclusion restriction concerns differential trends rather than levels: conditional on the controls and fixed effects, cities with different 2015 gender × age compositions would have followed parallel trends in antidepressant expenditure absent the uneven diffusion of social media across demographic groups. We assess this assumption in the *Robustness* analyses.

Results

Baseline. In the OLS specification, a one-hour increase in daily per-capita social media use is associated with a 5.6% increase in total per-capita antidepressant expenditure (Table 1, Panel A, column 1). The corresponding 2SLS estimate is larger, at 7.8% (Table 1, Panel B, column 1). The increase is concentrated in first-line medications, at 5.6% under OLS and 7.8% under 2SLS, compared with 0.6% for second-line medications in both specifications (Table 1, Panels A and B, columns 2 and 3). The reduced-form estimates are reported in Table S5.

Drug-by-drug evidence. The drug-by-drug analysis addresses three concerns: whether the aggregate effect masks heterogeneity across compounds, whether expenditure changes reflect price or product-mix shifts rather than quantity responses, and whether a small number of compounds drive the first-line versus second-line pattern. We therefore re-estimate the baseline OLS and 2SLS specifications separately for each of the 17 generic compounds, using log per-capita dosage as the outcome (Fig. 1, A and B; full estimates in Tables S6 and S7). Across both specifications, all eight first-line drugs show positive, statistically significant effects, whereas all second-line drugs show smaller effects, most clustering near zero. This pattern mirrors the aggregate first-line versus second-line results in Table 1.

Heterogeneity by platform type. We next examine whether the effect differs across platform types. We group the seven social media categories into two types: user-content-based platforms, built around user-generated content and sharing, and task-based platforms, oriented toward specific social or professional purposes (see Section S1.2 of the Supplementary Materials). We then re-estimate the baseline OLS and 2SLS specifications separately for each type.

The effect is concentrated in user-content-based platforms: a one-hour increase in daily per-capita use raises antidepressant expenditure by 5.7% under OLS and 7.8% under 2SLS. By contrast, task-based platforms show no detectable effect (−0.4% under OLS, 0.8% under 2SLS) (Fig. 2).

Heterogeneity by city characteristics. We next examine whether the effect varies with population vulnerability. We consider four city-level dimensions—age structure, educational attainment, employment rate, and marriage rate—measured with predetermined 2015 characteristics: the share of residents aged 15–34, the share of college-educated residents, the employment rate among residents aged 16 and above, and the share of married residents. For each characteristic, we split cities at the sample median and re-estimate the baseline OLS and 2SLS specifications separately for each side.

The effect is larger in cities with more vulnerable populations across all four dimensions: a one-hour increase in daily per-capita social media use raises per-capita antidepressant expenditure by 6.2% to 6.9% under OLS (Fig. S3) and 9.0% to 10.9% under 2SLS (Fig. 3) in cities with younger populations, lower educational attainment, lower employment rates, or lower marriage rates. In cities on the other side of each median, the effect remains positive but smaller (4.6% to 5.1% under OLS, 6.1% to 6.4% under 2SLS), with the 2SLS difference statistically significant for each of the four splits.

Placebo test using non-social-media app use. The estimated effect could in principle reflect broader mobile-app use rather than social media specifically. We test this by replacing social media use with per-capita non-social-media app use and its corresponding shift-share instrument. In sharp contrast to the main estimates, both OLS and 2SLS coefficients are statistically indistinguishable from zero (Table S8), supporting the interpretation that the effect is specific to social media exposure.

Robustness. We assess the robustness of the findings along two dimensions. First, we examine the identifying assumptions underlying the Bartik design by conducting a placebo test using later-period predicted social media use, testing for baseline imbalances, allowing for differential trends by predetermined city characteristics, including the demographic shares that construct the instrument, assessing whether identification is concentrated in a small number of demographic cells, and reconstructing the instrument using nationwide leave-one-out shifts (Tables S9–S13).

These checks should mitigate the concern that a time-varying unobserved confounder correlated with both the Bartik shares and shifts drives the results. In case any concern remains, we present three additional pieces of evidence. First, such a confounder would also produce a detectable effect

for task-based platforms, yet none appears (Fig. 2). Second, we include city–year fixed effects—a stringent specification that removes any confounder operating at the city–year level, at the cost of discarding the useful year-to-year identifying variation and leaving identification to month-to-month variation within each city–year. The estimated effect nevertheless remains positive and statistically significant (Table S14, column 6). Third, we directly address two salient potential confounders. Excluding the COVID period or controlling for citywide lockdowns leaves the estimates unchanged (Table S15), as does controlling for city–month indices of public perceptions of unemployment and recession (Table S16).

Second, we evaluate the sensitivity of the estimates to population weighting, sample restrictions excluding major public holidays or the four municipalities, province-level clustering, and city-specific linear trends (Table S14).

Mechanisms

Two channels can plausibly carry social media exposure to antidepressant demand: *time displacement* and *social perceptions*. We test each channel using individual-level microdata from the Chinese General Social Survey (CGSS). For each candidate measure—a time-use category or a perception index—we ask whether it correlates with self-reported depression, and whether it correlates with social media use in the respondent's city-by-gender-by-age cell.

Time displacement. Time is finite, so additional social media use must crowd out other activities; if the displaced activities are protective—and offline rest, in-person social interaction, and exercise have been linked to lower depression risk (*30–33*)—then exposure should raise depression. Two patterns should follow if the channel operates: offline activities should be negatively associated with depression, and social media use should displace those activities. Both hold in the data. Self-reported depression is negatively associated with every offline activity and positively associated with time spent online (Fig. 4A; Table S17, Panel A). More social media use is associated with more time online and less time on every offline activity—socializing, rest, exercising, meeting friends, shopping, visiting relatives, and studying (Fig. 4B; Table S17, Panel B).

Social perceptions. Social media exposes users to selectively curated content that overrepresents wealth, success, and conflict. Persistent exposure may distort perceptions of one's relative position

and the broader social environment toward greater inequality, lower trust, and bleaker outlooks—channels linked to depression (*34*). Two patterns should follow if the channel operates: the four perception measures should be positively associated with depression, and social media use should raise them. Both hold in the data. All four measures—perceived inequality, income underestimation, social distrust, and future pessimism—are positively associated with self-reported depression (Fig. 4C; Table S18, Panel A). More social media use is associated with higher levels on every measure (Fig. 4D; Table S18, Panel B).

Mechanisms and heterogeneity. Heterogeneity across platform types is consistent with both channels (Fig. 2). User-content-based platforms combine intensive engagement, which can displace offline activities, with repeated exposure to selectively curated content, which can distort social perceptions. Task-based platforms are used for specific purposes, limiting both engagement time and exposure to curated content, weakening both channels. Population heterogeneity may reflect differential vulnerability to the same two channels. Younger residents have less accumulated life experience, leaving them less able to contextualize curated content as filtered rather than representative. Lower educational attainment may leave residents less trained to recognize selection bias and curated presentation. Lower employment and marriage rates reduce work- and household-based social ties, leaving fewer offline interactions to buffer displaced time and counterbalance distorted comparisons. The larger effects observed in these cities (Fig. 3) are consistent with this interpretation. Together, the evidence suggests that social media's effects on antidepressant demand depend on both platform features and population vulnerability.

Awareness as an alternative explanation. One alternative interpretation is that social media exposure increases antidepressant demand by raising awareness of depression rather than worsening underlying mental health. First, controlling for a proxy for depression awareness (a city–month Baidu search index of depression-related keywords) barely changes the social media coefficient (Table S19). Second, social media exposure is associated with an increase in self-reported depression in the CGSS (Table S20), a symptom-frequency measure distinct from information seeking and treatment-seeking behavior. These results suggest that increased awareness is unlikely to be the main driver of the estimated effect. Further details are provided in Section S4.4 of the Supplementary Materials.

Discussion and Limitations

Social media raises antidepressant demand, but the effect differs sharply across platforms and populations. These findings advance the ongoing debate over whether and when social media harms mental health (*5, 7, 11, 16, 18*). Population-scale coverage, a clinical outcome, and cross-platform exposure move beyond an average effect and show that impact depends on both platform features and population vulnerability—helping to explain the mixed findings in prior work. Nationwide evidence from China, one of the world's most populous countries, also extends the literature beyond its concentration in high-income settings. These estimates should accordingly be read as population-level relationships: they capture how average demand differs across populations with different levels of exposure, rather than how any individual responds to their own use over time.

Two limitations point to individual-level extensions. Procurement records fall short of clinical diagnoses or prescription records, which would directly capture medically recognized conditions and prescribed treatment. Social media usage is measured at the city–month level, so the estimates capture population-level rather than individual-level exposure. Linking individual digital-trace records to diagnoses or prescriptions would identify individual-level effects, clarify which platforms affect which populations and conditions, and complement the population-scale design.

Table 1: **Social media use raises antidepressant expenditure.** This table reports estimates of the effect of daily per-capita social media usage (hours) on log per-capita antidepressant expenditure, separately for total antidepressant expenditure and expenditure on first-line and second-line medications. Panel A reports OLS estimates; Panel B reports 2SLS estimates using predicted social media use as an instrument for observed social media use. Predicted social media use is constructed using the shift-share (Bartik) design in equation (2). All specifications include city and year-month fixed effects and time-varying city-level controls. Standard errors clustered at the city level are reported in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

	Log per-capita antidepressant expenditure		
	Total	First-line	Second-line
	(1)	(2)	(3)
Panel A. OLS			
Social media usage time	0.0558***	0.0559***	0.0063***
	(0.0039)	(0.0040)	(0.0016)
Panel B. 2SLS			
Social media usage time	0.0775***	0.0780***	0.0060**
	(0.0053)	(0.0051)	(0.0027)
Kleibergen–Paap Wald F-statistic	559.9	559.9	559.9
City fixed effects	Yes	Yes	Yes
Year-month fixed effects	Yes	Yes	Yes
Controls	Yes	Yes	Yes
Observations	16,796	16,796	16,796



Figure 1: **Drug-by-drug evidence: social media exposure raises antidepressant demand, concentrated in first-line drugs.** (**A**) OLS and (**B**) 2SLS drug-by-drug estimates for log per-capita antidepressant dosage across 17 generic compounds. Vermilion circles denote first-line drugs; blue squares denote second-line drugs. The horizontal rule separates drug groups, the dashed line marks zero, and compounds are ordered within group by descending 2SLS estimate. All regressions include city and year-month fixed effects and time-varying city-level controls. Error bars indicate 95% confidence intervals based on standard errors clustered at the city level. The Kleibergen–Paap Wald F-statistic is 559.9 in panel B. $N = 16{,}796$ city–month observations per regression.

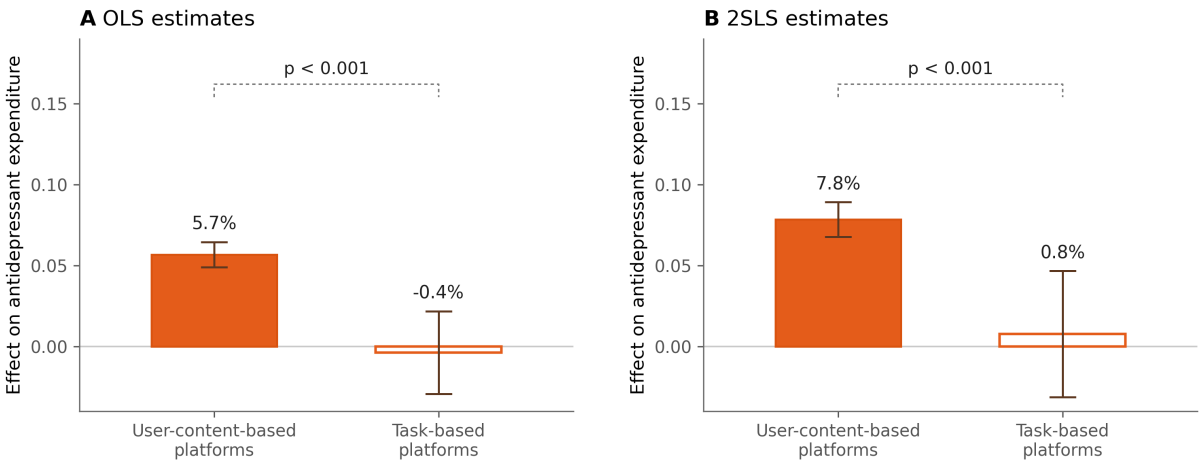


Figure 2: **Social media effects are concentrated in user-content-based platforms.** Estimates of the effect of a one-hour increase in daily per-capita usage of each platform type on log per-capita antidepressant expenditure, estimated separately for user-content-based and task-based platforms. (**A**) OLS estimates. (**B**) 2SLS estimates, each using a type-specific shift-share (Bartik) instrument constructed as described in Section S2.2 of the Supplementary Materials. The first-stage Kleibergen–Paap Wald F-statistics exceed 500 for both platform types. Filled bars denote user-content-based platforms, whereas hollow bars denote task-based platforms; bar labels report the estimated effects in percent. Error bars indicate 95% confidence intervals based on standard errors clustered at the city level. Brackets report p-values from tests of coefficient equality between the two platform types, using a fully interacted OLS specification in panel A and a fully interacted IV specification in panel B. All regressions include city and year-month fixed effects and time-varying city-level controls. $N = 16{,}796$ city–month observations per regression.

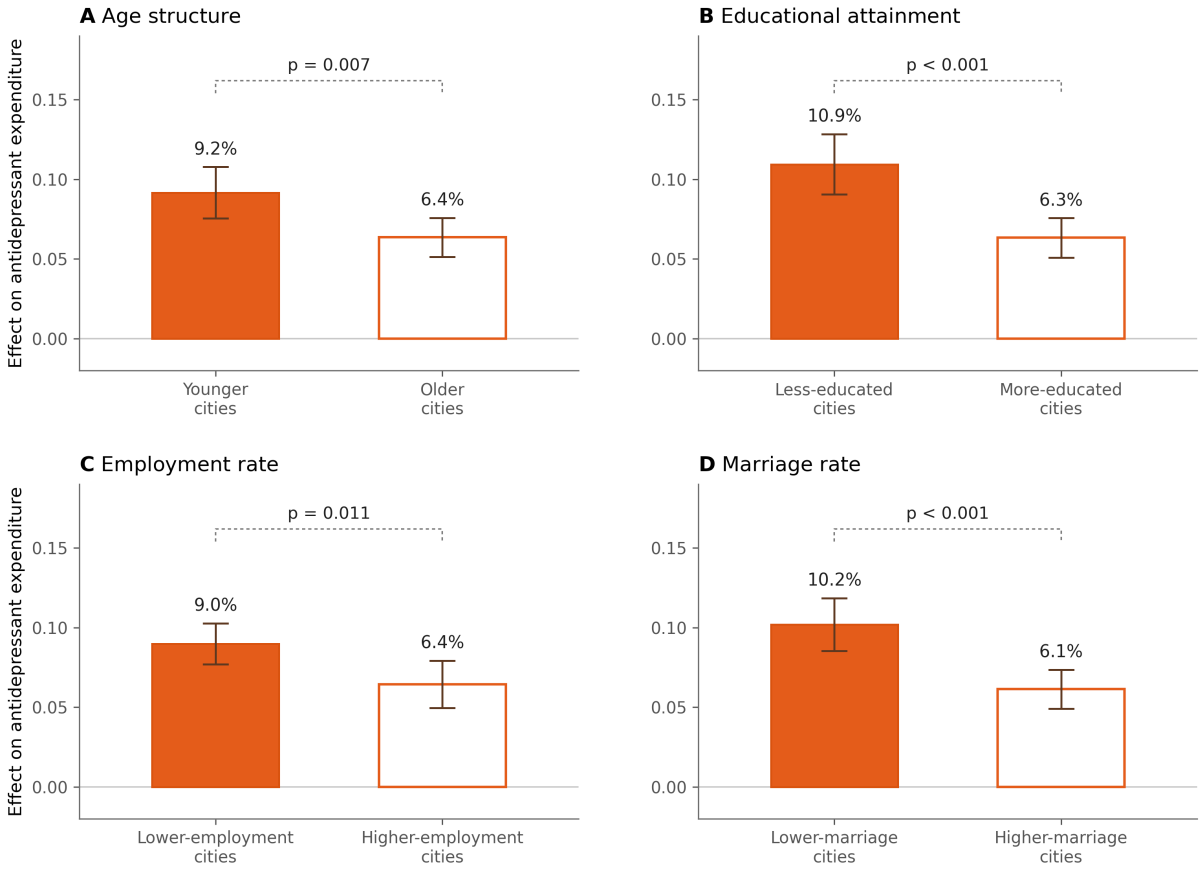


Figure 3: **Social media effects are stronger in more vulnerable populations.** IV estimates of the effect of a one-hour increase in daily per-capita social media usage on log per-capita antidepressant expenditure, splitting cities at the median of each 2015 characteristic. (**A**) Estimated effects by city age structure, comparing cities with above- versus below-median shares of residents aged 15–34. (**B**) Estimated effects by educational attainment, comparing cities with above- versus below-median shares of college-educated residents. (**C**) Estimated effects by employment rate, comparing cities with above- versus below-median employment rates among residents aged 16 and above. (**D**) Estimated effects by marriage rate, comparing cities with above- versus below-median marriage rates. Filled bars denote the more-vulnerable subsample, whereas hollow bars denote the comparison subsample. Error bars indicate 95% confidence intervals based on standard errors clustered at the city level. Brackets report p-values from tests of coefficient equality using a fully interacted IV specification. All regressions include city and year-month fixed effects and time-varying city-level controls. Each split divides the full panel of 16,796 city–month observations into the two subsamples.

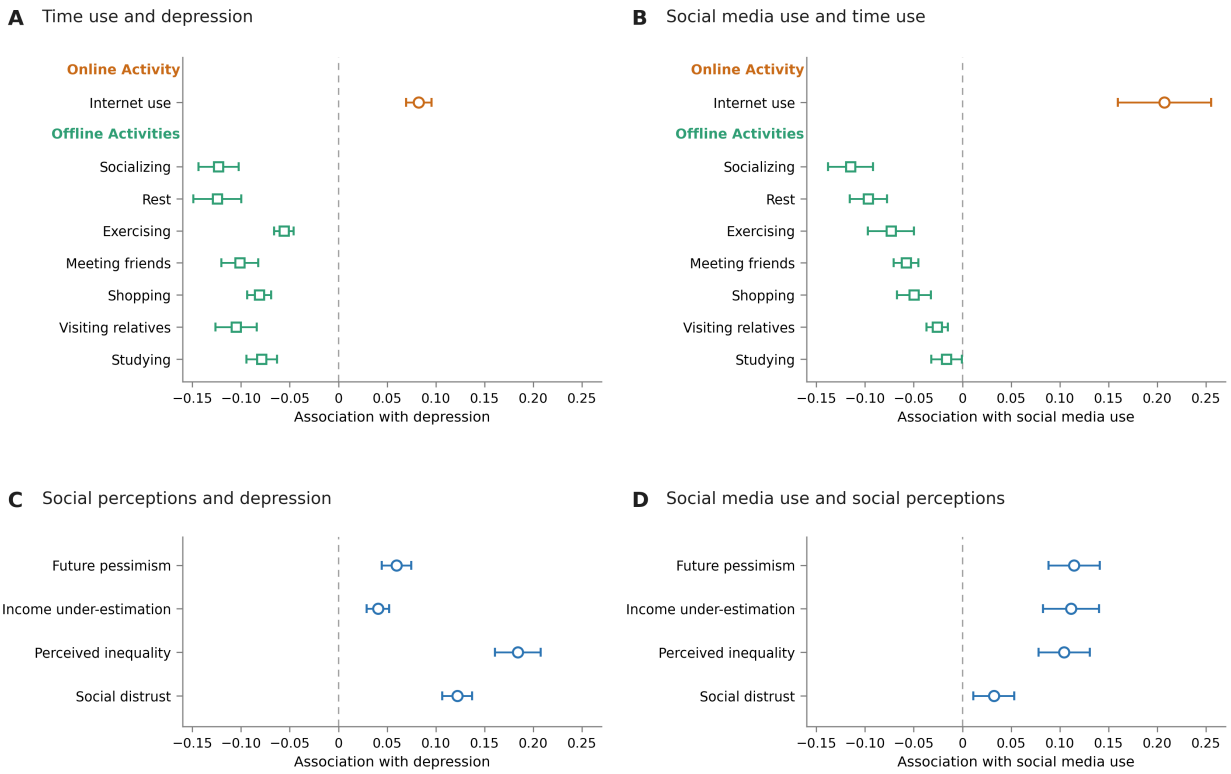


Figure 4: **Evidence consistent with two potential mechanisms linking social media exposure to antidepressant demand: time displacement and social perceptions.** Mechanism estimates from individual-level CGSS data. (**A**) OLS estimates of self-reported depression on each of eight time-use measures. Each activity measure is a five-point frequency scale (1 = never, 5 = every day). (**B**) OLS estimates of the association between social media use, measured for the respondent's corresponding city-by-gender-by-age cell, and each time-use measure. (**C**) OLS estimates of self-reported depression on each of four social-perception measures. (**D**) OLS estimates of the association between social media use, measured for the respondent's corresponding city-by-gender-by-age cell, and each social-perception measure. Each marker is a separate regression. In (A) and (B), vermilion circles denote online activity; green squares denote offline activities. All regressions include individual controls, city and year fixed effects, and time-varying city-level controls. Error bars indicate 95% confidence intervals based on standard errors clustered at the city level. Panels A and B use 34,311 respondents; panels C and D use 31,775 respondents.

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Acknowledgments

Author contributions: Conceptualization: H.C., J.H., Q.X. Methodology: H.C., J.H. Data curation: H.C., J.H., B.J., Q.X. Formal analysis: H.C., J.H. Resources: B.J., Q.X. Supervision: B.J. Visualization: J.H. Writing – original draft: H.C., J.H. Writing – review and editing: B.J., Q.X.

Competing interests: The authors declare no competing interests.

Data, Materials, and Software Availability: All analysis code, together with documentation, will be made available to editors and reviewers during peer review upon request and will be deposited in a persistent public repository upon acceptance. The Chinese General Social Survey (CGSS) data are available from the National Survey Research Center at Renmin University of China (https://cgss.ruc.edu.cn/). City-level socioeconomic indicators are from the China City Statistical Yearbook. Baidu search data are available from the Baidu Index platform (https://index.baidu.com/). The social media usage data were obtained under a commercial license from a Chinese internet analytics firm and cannot be redistributed by the authors; access requires a commercial license from

the provider. The antidepressant procurement records were obtained from China's National Drug Procurement Platform and cannot be redistributed by the authors under the platform's confidentiality policy; access requires authorization from the platform. China's 2015 1% National Population Sample Survey data are available through the National Bureau of Statistics–Peking University (NBS–PKU) Research Data Center at Peking University (https://microdata.stats.gov.cn/) but cannot be redistributed by the authors under the applicable data-use restrictions. Statistical analyses were performed using StataNow/MP version 19.5 (StataCorp LLC).

Supplementary materials

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Supplementary Materials for

Social media exposure raises antidepressant demand

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S1 Data sources and variable construction

This section describes the data sources and variable construction. We first introduce the two primary datasets, which comprise administrative records of antidepressant procurement and digital-trace data on social media use. We then describe the additional city-level socioeconomic data, population data, survey microdata, and Baidu search data. Finally, we explain how the variables are constructed.

S1.1 Medication procurement data

We obtain order-level administrative records of antidepressant procurement from China's National Drug Procurement Platform, which captures every purchase placed by medical institutions in mainland China. Each record specifies the purchasing institution, the medication, the date of procurement, the quantity, and the monetary value of the transaction.

We restrict attention to antidepressant medications under World Health Organization Anatomical Therapeutic Chemical (ATC) code N06A. The resulting sample covers 17 generic compounds across the major antidepressant classes represented in the procurement data. The data span January 2019 through April 2023 and comprise 2,512,878 transactions from 36,955 medical institutions across 323 prefecture-level cities.

Drug classification. We classify each compound as first-line or second-line according to its clinical treatment positioning. First-line medications are compounds that typically serve as initial pharmacological options for new or mild-to-moderate depressive symptoms. In our sample, this group includes the six SSRIs, duloxetine, and agomelatine. Second-line medications are compounds more often reserved for follow-up treatment or for complex, recurrent, or treatment-resistant cases. This group includes venlafaxine, milnacipran, mirtazapine, mianserin, reboxetine, amitriptyline, clomipramine, doxepin, and flupentixol–melitracen. The classification is fixed at the compound level before estimation and does not vary across cities or time. It should be interpreted as a clinical treatment-positioning measure rather than a patient-level diagnosis or a direct measure of disease severity. Table S1 lists each compound with its therapeutic class and assigned clinical line.

External validation. To validate that the procurement-based measure reflects the actual mental health of local populations, we link local antidepressant expenditure to respondent-level self-reported depressive symptoms from the Chinese General Social Survey (CGSS), a nationally

representative repeated cross-sectional survey conducted by the National Survey Research Center at Renmin University of China. The CGSS asks respondents how often they felt depressed during the preceding four weeks, with responses measured on a five-point scale from 1 (never) to 5 (always).

Using the CGSS waves that overlap with the procurement sample—the 2021 and 2023 waves—we assign each respondent the log of annual per-capita antidepressant expenditure for the respondent's city in the survey year. The relationship is strongly positive. Respondents in cities with higher log per-capita antidepressant expenditure at the time of the survey report more frequent depressive symptoms (slope = 0.414, $p < 0.001$; Fig. S1), supporting the use of antidepressant procurement as a population-level proxy for depression.

S1.2 Social media data

We measure social media usage with population-level digital-trace data covering the full set of nearly 600 social media applications operating in the Chinese market, grouped into seven categories: short-video (e.g., Douyin), microblog (e.g., Weibo), community and forum (e.g., Xiaohongshu and Baidu Tieba), instant messaging (e.g., WeChat), dating, professional networking, and other social-media apps. The data come from a leading Chinese internet analytics provider that, in collaboration with the country's three major telecommunications carriers (China Mobile, China Unicom, and China Telecom), uses deep packet inspection at the network level to passively record real-time mobile-application activity across more than 83% of mobile devices nationwide.

The digital-trace data provide aggregate time spent on social media applications, organized into city × month × gender × age cells for all 323 cities, with age groups as defined by the data provider: under 18, 18–24, 25–30, 31–35, 36–40, 41–45, 46–50, 51–60, and above 60. Each cell reports the total time spent on each social media application across all devices in that cell. We aggregate these cell-level time-use measures to a daily per-capita social media usage variable in hours (see *Variable construction* below) and use it as our baseline exposure measure. The provider's records extend back to 2017; the baseline panel uses January 2019 through April 2023 to match the procurement data. For a robustness analysis, the provider also supplies aggregate time spent on non-social-media mobile applications using the same cell structure.

Daily per-capita usage averages 1.84 hours in the baseline city–month panel (Table S2), rising

from 1.46 hours in 2019 to 2.14 hours in January–April 2023, with the short-video category accounting for the largest share of measured time and usage concentrated among users aged 18–40; the full city–month distribution, year-by-year trend, app-category mix, and demographic breakdown are reported in Fig. S2. A one-hour increase in daily per-capita usage therefore corresponds to roughly 54% of the sample mean.

Platform classification. We classify the seven social media categories into two broad platform types. The first type consists of user-content-based platforms, where users primarily consume, produce, and share user-generated content. This type includes four social media categories. First, the short-video category is centered on short-form video viewing and sharing, such as Douyin. Second, the microblog category is organized around public posting, reposting, commenting, and following, such as Weibo. Third, the community and forum category is centered on discussion communities, interest groups, and forum-style interactions, such as Xiaohongshu, Baidu Tieba, Douban, and Zhihu. Fourth, the instant messaging category includes apps such as WeChat and QQ. In China, these apps are not limited to point-to-point communication; they also include social-feed, group-sharing, public-account, and content-sharing functions, such as WeChat Moments and Qzone. We therefore classify them as user-content-based platforms.

The second type consists of task-based platforms, which are oriented toward specific social or professional purposes rather than the broad consumption and circulation of user-generated content. This type includes three social media categories. First, the dating category supports dating and relationship matching, such as Momo. Second, the professional networking category supports professional networking and career-related interactions, such as LinkedIn. Third, we classify the provider's other social-media category as task-based. Although these apps serve heterogeneous functions, including campus services, hobby participation, and local exchange, they are generally oriented toward specific, goal-directed activities. Because the category contains relatively few apps spread across many narrow functions, we retain it as a single category rather than subdivide it into small, weakly defined groups.

S1.3 City-level socioeconomic and population data

We draw annual city-level socioeconomic data from the China City Statistical Yearbook, including resident population, GDP per capita, fiscal expenditure per capita, secondary and tertiary industry shares, and hospital beds and doctors per 1,000 residents. We additionally use China's 2015 1% National Population Sample Survey to construct each city's predetermined population shares across gender $\times$ age groups and to measure four city characteristics used in the heterogeneity analyses: the share of residents aged 15–34, the share of college-educated residents, the employment rate among residents aged 16 and above, and the share of married residents.

S1.4 Chinese General Social Survey

The mechanism analysis uses microdata from the Chinese General Social Survey (CGSS), a nationally representative repeated cross-sectional survey of adults conducted by the National Survey Research Center at Renmin University of China. The CGSS uses a multistage stratified probability sampling design. We use the 2017, 2018, 2021, and 2023 waves, for which corresponding digital-trace exposure measures are available. The survey provides respondent-level measures of self-reported depression, time use, and social perceptions. The time-use measures cover internet use, offline socializing, rest, offline exercising, meeting friends, offline shopping, visiting relatives, and offline studying. The social-perception measures include future pessimism, income underestimation, perceived inequality, and social distrust. The time-use analysis uses 34,311 respondents, and the social-perception analysis uses 31,775 respondents. The digital-trace app data are available from 2017 onward, while the baseline procurement-linked panel starts in 2019 because the procurement data begin then.

S1.5 Baidu search data

We collect search-volume data from Baidu, China's dominant online search engine. The data are measured at the city–month level and cover all 323 cities from January 2019 through April 2023. For depression-related searches, the keyword set contains four groups: general depression and mental-health terms; symptom and distress terms; treatment and care-related terms; and antidepressant drug-name terms corresponding to the compounds listed in Table S1. For the robustness analysis,

we additionally collect search volumes for unemployment-related terms, including employment, unemployment, job search, recruitment, wages, and layoffs, and recession-related terms, including business closures, bankruptcy, debt, housing-market declines, consumption downgrading, inflation, and deflation. The construction of the depression-related Baidu search index, the unemployment perception index, and the recession perception index is described below.

S1.6 Variable construction

City–month panel and primary variables. We aggregate the procurement and digital-trace data to a balanced city–month panel and link them by city and calendar month. From the procurement records, we assign each order to a prefecture-level city and calendar month using the purchasing institution's location and transaction date. We sum order values within each city–month cell, divide by the city's resident population, and take the natural logarithm, yielding log per-capita antidepressant expenditure in RMB. We construct this outcome for total antidepressant expenditure and separately for first-line and second-line medications.

To assess whether changes in antidepressant expenditure reflect greater antidepressant demand rather than price or product-mix shifts, we also construct per-capita dosage in milligrams for each of the 17 generic compounds. We convert procurement quantities to milligrams using the package-level dosage information in each transaction, sum within each city–month–compound cell, divide by the city's resident population, and take the natural logarithm. The classification of the 17 compounds is described in Section S1.1 and reported in Table S1.

For each city–month, we sum digital-trace usage time across social media applications and demographic cells, then divide by the city's resident population and by the number of days in the month, yielding daily per-capita social media usage in hours. We apply the same procedure separately to the two platform types—user-content-based and task-based platforms (see *Platform classification* in Section S1.2)—to construct the type-specific exposure measures. Per-capita usage of non-social-media mobile applications, used in the placebo analysis, is constructed analogously.

Merging the procurement, social media, and city-level data yields a balanced panel of 323 cities observed from January 2019 through April 2023, comprising 16,796 city–month observations.

Demographic shares and city characteristics. Using China's 2015 1% National Population Sample

Survey, we compute each city's predetermined population shares across the gender × age cells defined in Section S1.2. These shares enter the shift-share instrument described in Section S2.2.

We also construct four predetermined city characteristics measured in 2015: the share of residents aged 15–34, the share of college-educated residents, the employment rate among residents aged 16 and above, and the share of married residents. For each heterogeneity analysis, we split cities at the sample median of the corresponding characteristic.

CGSS variables. We construct three sets of respondent-level variables. First, self-reported depression is measured by a five-point item asking how often the respondent felt depressed during the preceding four weeks, ranging from 1 (never) to 5 (always). Second, the time-use measures record the frequency of engagement in eight activity categories, each measured on a five-point scale from 1 (never) to 5 (every day): internet use, offline socializing, rest, offline exercising, meeting friends, offline shopping, visiting relatives, and offline studying.

Third, the social-perception measures comprise future pessimism, income underestimation, perceived inequality, and social distrust. Future pessimism is measured as the respondent's current social-status rating minus the expected social-status rating ten years from now, both ranging from 1 to 10; higher values indicate a larger expected decline in social status and thus greater pessimism about the future. Income underestimation is measured as the respondent's actual income quintile minus the self-assessed income quintile, both ranging from 1 to 5; higher values indicate greater underestimation of the respondent's relative income position. Perceived inequality (*society is unequal*) and social distrust (*others would take advantage of the respondent*) are each measured on a five-point scale, with higher values indicating greater perceived inequality and greater social distrust, respectively.

For the mechanism analyses, we assign each respondent the daily per-capita social media usage measure for the respondent's corresponding city-by-gender-by-age cell at the time of the survey.

Baidu search indices. For the three keyword sets described above, we construct the depression-related Baidu search index, the unemployment perception index, and the recession perception index. For each index, we apply principal component analysis (PCA) to the corresponding keyword-level city–month search volumes over the full sample and use the first principal component as the city–month index. We standardize each index over the city–month sample to have mean zero and unit standard deviation. Higher values indicate greater local search activity related to the corresponding

topic.

Summary statistics are reported in Tables S2 and S3.

S2 Empirical specifications and identification

S2.1 Baseline specification

We estimate the effect of social media use on per-capita antidepressant expenditure with the baseline city–month specification:

$$\ln(Y_{ct}) \;=\; \beta\,\mathrm{SMUsage}_{ct} \;+\; X_{ct}'\gamma \;+\; \mu_c \;+\; \lambda_t \;+\; \varepsilon_{ct}, \tag{S1}$$

where Y_{ct} is per-capita antidepressant expenditure (total, first-line, or second-line) for city c in year-month t, $\mathrm{SMUsage}_{ct}$ is daily per-capita social media usage in hours, X_{ct} is the vector of time-varying city-level controls, and μ_c and λ_t are city and year-month fixed effects. The coefficient β is the percentage change in per-capita antidepressant expenditure per one-hour increase in daily social media use.

S2.2 Bartik instrument

OLS estimates of equation (S1) may be biased by unobserved local shocks driving both social media use and antidepressant procurement, or by reverse causality if worse mental health raises social media use. We address these concerns with a shift-share (Bartik) instrument combining pre-determined city demographic composition with leave-one-out usage among the same demographic cells in other cities of the same province.

Let p denote the province containing city c, and let $d = (g, a)$ index gender × age cells, with age groups as described in Section S1.2 (under 18, 18–24, 25–30, 31–35, 36–40, 41–45, 46–50, 51–60, and above 60). The share s_{cd}^{2015} is city c's 2015 population share in cell d, constructed from China's 2015 1% National Population Sample Survey; the shift $\overline{\mathrm{SMUsage}}_{p,-c,t,d}$ is the average daily usage of cell d in year-month t across other cities of province p, excluding c. The instrument (i.e., predicted social media use) is defined as follows:

$$Z_{ct} \;=\; \sum_{d} s_{cd}^{2015}\,\overline{\mathrm{SMUsage}}_{p,-c,t,d}. \tag{S2}$$

The instrument varies across cities through predetermined demographic composition and over time through the leave-one-out shifts; the focal city's own usage is excluded from the shift, so Z_{ct} is not mechanically driven by contemporaneous local use.

We estimate β by two-stage least squares as in equations (3) and (4) of the main text; both stages carry the same city and year-month fixed effects and time-varying controls as equation (S1). The full first-stage estimates are reported in Table S4: predicted social media use enters with a positive, precisely estimated coefficient, and the Kleibergen–Paap Wald F-statistic exceeds 500 in the baseline specification, well above conventional thresholds for weak instruments.

For Beijing, Shanghai, Tianjin, and Chongqing, which have no other cities within the same provincial-level unit, we construct the shift using social media usage among the same demographic group in all other cities nationwide, excluding the focal municipality.

For the platform-type heterogeneity analysis (Fig. 2), we construct type-specific instruments analogously. For each platform type k—user-content-based or task-based (see Section S1.2)—the shift is the leave-one-out average daily usage of platform type k among cell d in other cities of the same province, and the instrument is $Z^k_{ct} = \sum_d s^{2015}_{cd} \, \overline{\text{SMUsage}}^{\,k}_{p,-c,t,d}$. The demographic shares are identical to the baseline design; only the shift varies by platform type. Each type-specific 2SLS regression then instruments usage of that platform type with the corresponding Z^k_{ct}, with the same fixed effects and controls as the baseline specification.

S2.3 Identifying assumptions

We invoke the shares-based interpretation of shift-share designs (*1*): the identifying variation comes from each city's predetermined 2015 gender × age composition, and the 2SLS estimator is numerically equivalent to a generalized method of moments estimator that uses these shares as instruments. The shares are plausibly orthogonal to antidepressant-demand trends because they are fixed at their 2015 values—four years before the sample opens and before the social media diffusion we study—and reflect slow-moving demographic structure shaped by historical fertility, cohort size, and migration rather than current conditions in the local mental-health-care market.

Identification requires that, conditional on the controls and fixed effects, the interaction between predetermined demographic composition and leave-one-out demographic usage affects antidepres-

sant procurement only through local social media exposure. This assumption fails if predetermined shares predict differential trends in antidepressant demand for reasons unrelated to social media, or if the leave-one-out shifts from other cities in the same province capture shocks shared within the province that directly affect the focal city. We assess these concerns through supplementary checks reported in Section S3.

S3 Robustness analyses

We probe the robustness of the main findings along two dimensions: the validity of the Bartik instrument, with particular attention to the identifying concerns outlined in Section S2.3; and the sensitivity of the estimates to alternative empirical specifications and sample restrictions.

S3.1 Validity of the Bartik instrument

Following the shares-based interpretation in the existing literature (*1*), we assess the exogeneity of the predetermined demographic shares by examining whether they predict changes in antidepressant demand through channels other than social media.

Placebo test using later-period predicted social media use. We conduct a placebo test to assess whether the Bartik instrument captures pre-existing differences in antidepressant-demand trends. Because antidepressant procurement data are available only from 2019 onward, no pre-sample outcomes are available. We therefore carry out a standard two-sub-period pre-trends test: the earlier period (2019, 2020, and 2021) provides the outcomes, and the later period (2022) provides the instrument. Specifically, we construct predicted social media use for each calendar month of 2022 and match it to antidepressant expenditure in the corresponding calendar month of 2019, 2020, or 2021. Under this design, a persistent unobserved factor jointly shaping predicted social media use and antidepressant demand would produce a systematic correlation between later predicted use and earlier outcomes. The estimated coefficients are 0.0085, -0.0020, and 0.0060, none statistically significant (Table S9). This null pattern supports the identification strategy.

Allowing differential trends. A central concern for the exclusion restriction is that cities with different predetermined characteristics may have followed different antidepressant-demand trajectories during the sample period for reasons unrelated to social media. We address this concern by al-

lowing cities to follow different trajectories along each predetermined dimension, interacting it with year-month fixed effects. The most demanding specification interacts year-month fixed effects with the 2015 gender × age population shares that construct the instrument, allowing every demographic group its own month-by-month trajectory. This removes any confounder operating through demographic-specific time trends, at the cost of absorbing much of the variation that identifies the baseline effect. The estimate nevertheless remains close to the baseline at 0.0833 (Table S10, column 1). We also allow separate trajectories by the share of college-educated residents, the employment rate, the share of married residents, population, GDP per capita, and mobile penetration, one at a time (0.0754 to 0.0780; Table S10, columns 2 to 7) and all seven sets of interactions jointly (0.0830; Table S10, column 8). All estimates remain statistically significant at the 1% level. This stability reduces concern that differential trends associated with predetermined demographic composition, socioeconomic conditions, or technology penetration drive the estimated effect.

Balance on predetermined characteristics. A related concern is that predicted social media use may be systematically correlated with pre-existing city characteristics that independently shape antidepressant demand. We therefore test its association with city characteristics measured in 2015, including GDP per capita, fiscal expenditure per capita, the shares of secondary and tertiary industry, hospital beds and doctors per 1,000 residents. As shown in Table S11, we find no significant association between predicted social media use and any of these characteristics, either individually or jointly. This broad balance reduces concern that the instrument systematically loads on pre-existing differences in local economic development, industrial structure, or medical capacity that could independently shape antidepressant demand, thereby providing further support for the validity of the Bartik instrument.

Rotemberg weights. We next examine whether the identifying variation is concentrated in a small number of gender × age cells. As shown in Table S12, the largest Rotemberg weight is 9.5%, and the five largest weights together account for 45.7%. The corresponding just-identified IV estimates, which range from 0.080 to 0.082, are close to the baseline estimate. These diagnostics indicate that identification is not concentrated in a single demographic cell and that the main estimate does not hinge on one narrow gender × age comparison.

Alternative instrument. We also reconstruct the instrument using leave-one-out social media usage among the same demographic groups nationwide rather than within the same province. This

alternative construction reduces concern that the baseline instrument captures shocks shared by cities within the same province. As shown in Table S13, it yields an estimate of 0.0740 ($p < 0.01$), close to the baseline estimate of 0.0775, with a Kleibergen–Paap Wald F-statistic of 20.8.

Confounders potentially correlated with the shares and the shifts. Despite the checks above, one might still worry that an unobserved confounder correlated with both the shares and the shifts—a pandemic or a labor-market shock, for example—drives the results. We take three steps to address this possibility. First, such a confounder would contaminate the estimate for task-based platforms as well. The type-specific instruments for user-content-based and task-based platforms are built on the same 2015 gender × age shares and differ only in the platform-type-specific shifts (Section S2.2); moreover, usage of the two platform types diffused over the same period across the same demographic groups, so the type-specific shifts co-move. A confounder correlated with the shares and the shifts would therefore bias both platform-type estimates. Both type-specific first stages are strong, so the null estimate for task-based platforms does not reflect a weak instrument. Yet task-based platforms show no detectable effect (Fig. 2).

City–year fixed effects. Second, we include city–year fixed effects, a stringent specification that removes any confounder operating at the city–year level, at the cost of absorbing the between-year variation in social media diffusion that identifies the baseline effect and leaving identification to month-to-month variation within each city–year. Because it discards useful identifying variation, we use it as a robustness check rather than as the baseline. The estimated effect nevertheless remains positive and statistically significant: the 2SLS coefficient is 0.0498 (SE = 0.0054, $p < 0.001$), and the first-stage Kleibergen–Paap Wald F-statistic is 326.3 (Table S14, column 6).

Third, we examine directly two salient potential confounders—COVID-19 and local economic conditions—both of which plausibly varied with age and evolved over the sample period. As we show next, neither accounts for the estimated effect.

The COVID-19 pandemic. We address the potential concern raised by COVID-19 in two ways. First, we control for citywide lockdowns, using an indicator for city–months under citywide lockdown, following the existing literature (*2*). The coefficient on social media usage is nearly unchanged at 0.0774 (Table S15, column 1). Second, we exclude the COVID-period years from 2020 to 2022 altogether. The resulting estimate remains similar to the baseline at 0.0748 and is statistically significant at the 1% level (Table S15, column 2).

Local economic conditions. Using search-volume data from Baidu, we construct city–month indices capturing public perceptions of unemployment and recession (see Section S1.6). Each index is constructed as the first principal component of the corresponding keyword-level search volumes and standardized to have a mean of zero and a standard deviation of one. As shown in Table S16, predicted social media use is not significantly associated with either the unemployment perception index or the recession perception index, with estimated coefficients of 0.0001 and −0.0001, respectively. Moreover, controlling for either index leaves the coefficient on social media usage unchanged at 0.0775. These results reduce concern that changes in local economic conditions drive the estimated effect.

S3.2 Alternative empirical specifications and sample restrictions

We examine whether the findings are robust to alternative empirical specifications and sample restrictions.

Weighting by city population. Population weighting yields a larger estimate of 0.0953, which remains statistically significant at the 1% level, indicating that the baseline result is not disproportionately driven by smaller cities (Table S14, column 1).

Excluding major public holidays. The estimate also remains similar after excluding months containing major public holidays, at 0.0828 (Table S14, column 2), indicating that holiday-related changes in social media use or healthcare procurement do not account for the findings.

Excluding the four municipalities. Because Beijing, Shanghai, Tianjin, and Chongqing are provincial-level units with no other cities within the same province, we also re-estimate the model after excluding these four municipalities. The resulting estimate is 0.0770, nearly identical to the baseline estimate (Table S14, column 3).

Clustering standard errors by province. Because the instrument uses leave-one-out usage among cities in the same province, shocks shared within a province may correlate residuals across cities in ways that city-level clustering does not accommodate. We therefore cluster at the province level. The point estimate is unchanged at 0.0775; the standard error rises from 0.0053 to 0.0121, and the estimate remains statistically significant at the 1% level (Table S14, column 4).

Allowing city-specific linear trends. Finally, we augment the baseline specification with city-specific

linear time trends, allowing each city to follow its own smooth trajectory in antidepressant expenditure. The resulting estimate is 0.0908 and remains statistically significant at the 1% level (Table S14, column 5), indicating that the baseline result is not driven by smooth city-specific trajectories.

S3.3 Placebo test using non-social-media app use

To test whether the estimated effect reflects broader mobile-app use rather than social media specifically, we replace social media use with per-capita non-social-media app use and construct its corresponding shift-share instrument. In sharp contrast to the main estimates, both OLS and 2SLS coefficients are statistically indistinguishable from zero: 0.0011 (SE = 0.0028, $p = 0.696$) under OLS and 0.0013 (SE = 0.0043, $p = 0.760$) under 2SLS, with a strong first stage (Kleibergen–Paap Wald F-statistic of 111.4; Table S8). This pattern supports the interpretation that the effect is specific to social media exposure.

S4 Mechanisms and an alternative explanation

We examine two potential behavioral mechanisms linking social media exposure to antidepressant demand: time displacement and social perceptions. We then consider increased depression awareness as an alternative explanation.

S4.1 Empirical framework

We use individual-level CGSS microdata to examine two potential behavioral mechanisms consistent with the city-level procurement response: time displacement and social perceptions. For each mechanism measure M_{ict}, we estimate

$$D_{ict} = \theta M_{ict} + W_{ict}'\eta + X_{ct}'\gamma + \mu_c + \lambda_t + \xi_{ict}, \tag{S3}$$

which relates the depression score D_{ict} to M_{ict} by OLS, and

$$M_{ict} = \delta \mathrm{SMUsage}_{c,t,d(i)} + W_{ict}'\eta + X_{ct}'\gamma + \mu_c + \lambda_t + \nu_{ict}, \tag{S4}$$

which relates M_{ict} to social media use in the respondent's corresponding city-by-gender-by-age cell by OLS. Both specifications are descriptive: equation (S3) links each candidate mechanism to

depression, and equation (S4) links social media exposure in the corresponding city-by-gender-by-age cell to that mechanism. Both are estimated separately for each of the eight time-use measures and four social-perception measures, yielding the 24 coefficients in Fig. 4. In equations (S3) and (S4), i indexes respondents and t indexes the survey year, so λ_t denotes survey-year fixed effects. The individual control vector W_{ict} includes gender, age, rural *hukou* status, minority ethnicity, education, and employment; city-level controls and fixed effects follow the baseline specification. Tables S17 and S18 report the underlying estimates.

Standard errors are clustered at the city level throughout, accommodating arbitrary within-city correlation and heteroskedasticity across cities. Significance levels are denoted by $^{*}p < 0.10$, $^{**}p < 0.05$, and $^{***}p < 0.01$.

S4.2 Time displacement

One potential mechanism linking social media use to depression is time displacement. By increasing time spent online, social media use may crowd out rest, exercise, in-person social interaction, and other offline activities that may support mental health. We examine this mechanism using detailed CGSS measures of respondents' time use. Specifically, the CGSS asks respondents how frequently they engage in eight activities: internet use, offline socializing, rest, offline exercising, meeting friends, offline shopping, visiting relatives, and offline studying. Each activity is measured on a five-point frequency scale ranging from 1 (never) to 5 (every day). Additional details on variable construction are provided in Section S1.6.

We first estimate equation (S3) to examine how each time-use measure is associated with self-reported depression. The results are shown in Fig. 4A, with the underlying estimates reported in Table S17, Panel A. We find that internet use is positively associated with self-reported depression (coefficient = 0.082, $p < 0.01$), whereas all seven offline activities are negatively associated with self-reported depression. The largest negative associations are observed for rest (−0.124), offline socializing (−0.123), visiting relatives (−0.105), and meeting friends (−0.101); all are statistically significant at the 1% level.

We then estimate equation (S4) to examine how social media use is associated with each time-use measure. The results are shown in Fig. 4B, with the underlying estimates reported in Table S17,

Panel B. On the five-point frequency scale, a one-hour increase in daily social media use is associated with a 0.21-point increase in the frequency of internet use ($p < 0.01$). The corresponding estimates are -0.12 for offline socializing, -0.10 for rest, -0.07 for exercising, -0.06 for meeting friends, -0.05 for offline shopping, and -0.03 for visiting relatives (all $p < 0.01$), as well as -0.02 for offline studying ($p < 0.05$).

Taken together, these results suggest a shift in time use away from offline activities and toward internet use—from a pattern associated with lower depression to one associated with higher depression—consistent with time displacement as a mechanism linking social media exposure to antidepressant demand.

S4.3 Social perceptions

A second potential mechanism linking social media use to depression operates through social perceptions. Social media exposes users to selectively curated depictions of wealth, success, conflict, and other salient experiences. Repeated exposure may heighten perceived inequality, lead individuals to underestimate their relative income position, weaken social trust, and foster pessimism about the future, each of which may adversely affect mental health (*3*). To examine this mechanism, we draw on four CGSS measures of respondents' social perceptions. Future pessimism is defined as the respondent's current social-status rating minus the expected rating ten years later, so higher values indicate a larger expected decline. Income underestimation is defined as the respondent's actual income quintile minus the self-assessed income quintile, so higher values indicate greater underestimation of relative income. Perceived inequality and social distrust are measured on five-point scales, with higher values indicating greater perceived inequality and stronger distrust of others. Additional details on variable construction are provided in Section S1.6.

We first estimate equation (S3) to examine how each social-perception measure is associated with self-reported depression. The results are shown in Fig. 4C, with the underlying estimates reported in Table S18, Panel A. All four measures are positively associated with self-reported depression. The estimated coefficients are 0.060 for future pessimism, 0.041 for income underestimation, 0.184 for perceived inequality, and 0.122 for social distrust; all are statistically significant at the 1% level.

We next estimate equation (S4) to examine how social media use is associated with each social-perception measure. The results are shown in Fig. 4D, with the underlying estimates reported in Table S18, Panel B. A one-hour increase in daily social media use is associated with increases of 0.12 in future pessimism, 0.11 in income underestimation, 0.10 in perceived inequality, and 0.03 in social distrust; all four estimates are statistically significant at the 1% level.

These results suggest that greater social media use is associated with a shift in perceptions of one's relative position and social environment, with each dimension moving in a direction associated with higher depression—consistent with social perceptions as another mechanism linking social media exposure to antidepressant demand.

S4.4 Awareness as an alternative explanation

One alternative interpretation is that social media exposure increases antidepressant demand by raising awareness of depression, rather than by worsening underlying mental health. We evaluate this alternative in two ways. First, we construct a proxy for depression awareness. Specifically, we use search-volume data from Baidu, China's dominant online search engine, for depression-related keywords and summarize them as a city–month Baidu search index (see Section S1.6). Because greater awareness should lead individuals to search more actively for depression-related information, we interpret higher values of this index as a proxy for greater depression awareness.

With this index constructed, we first find that the depression-related Baidu search index is positively associated with antidepressant expenditure (Table S19, column 1), consistent with the idea that greater awareness of depression is related to higher antidepressant demand. However, if increased awareness were the primary driver of the estimated social media effect, controlling for the Baidu search index should substantially reduce the estimated coefficient on social media exposure and make it statistically insignificant. Instead, when the Baidu search index is added, the coefficient on social media exposure remains positive and highly significant in both the OLS and 2SLS estimates (Table S19, columns 2 and 3), and the magnitudes are very close to the corresponding baseline estimates in Table 1, Panel A, column 1, and Panel B, column 1. This suggests that increased awareness does not appear to be the main driver of the increased antidepressant expenditure.

Second, we examine self-reported depression in survey data. If the baseline results were driven

mainly by greater awareness rather than worsening mental health, we would not expect social media exposure to increase self-reported depression. Yet the results show that social media exposure is also associated with an increase in self-reported depression (Table S20). Taken together, both pieces of evidence suggest that increased awareness is unlikely to be the primary driver for the main effect of social media exposure on antidepressant expenditure.

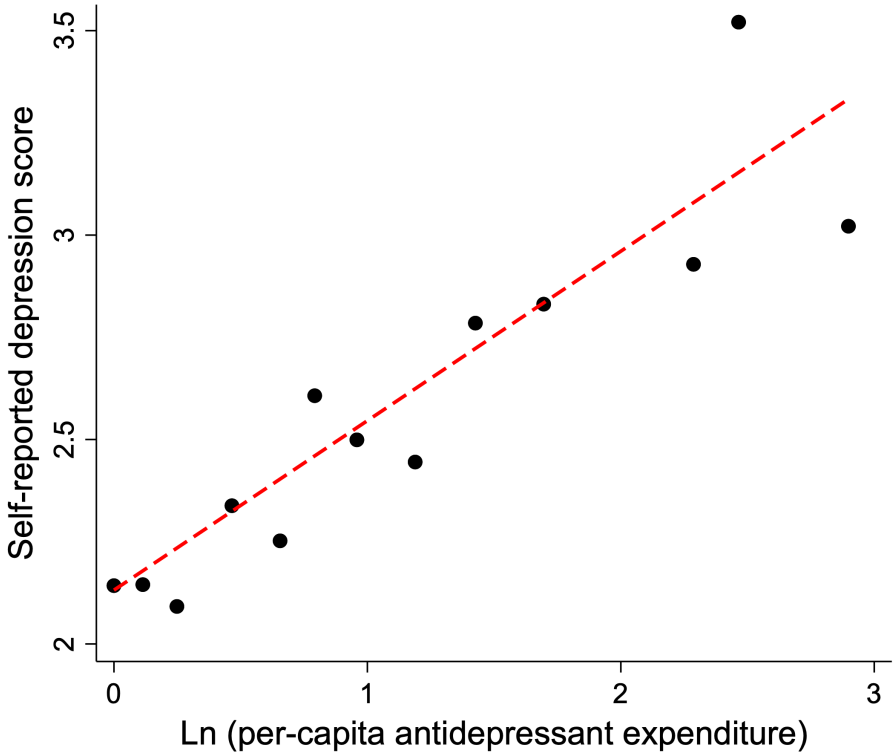


Figure S1: **External validation of the antidepressant procurement measure: positive correlation with self-reported depression from the Chinese General Social Survey.** This figure provides external validation of the administrative procurement-based measure of antidepressant demand used throughout the paper, interpreted as a population-level proxy for depression. The horizontal axis plots the natural logarithm of city–year per-capita antidepressant expenditure, constructed from the procurement data described in Section S1.1. The vertical axis plots CGSS respondents' self-reported depression score; higher values indicate more frequent self-reported depressive feelings. The scatter shows binned means of respondent-level observations, with bins constructed from the original log expenditure variable; the dashed red line reports the linear best fit estimated from the original respondent-level data. The relationship is positive and statistically significant: respondents in city–years with higher per-capita antidepressant expenditure report higher depression scores in the survey data. This pattern supports the interpretation of administrative antidepressant procurement as a meaningful population-level proxy for depression.

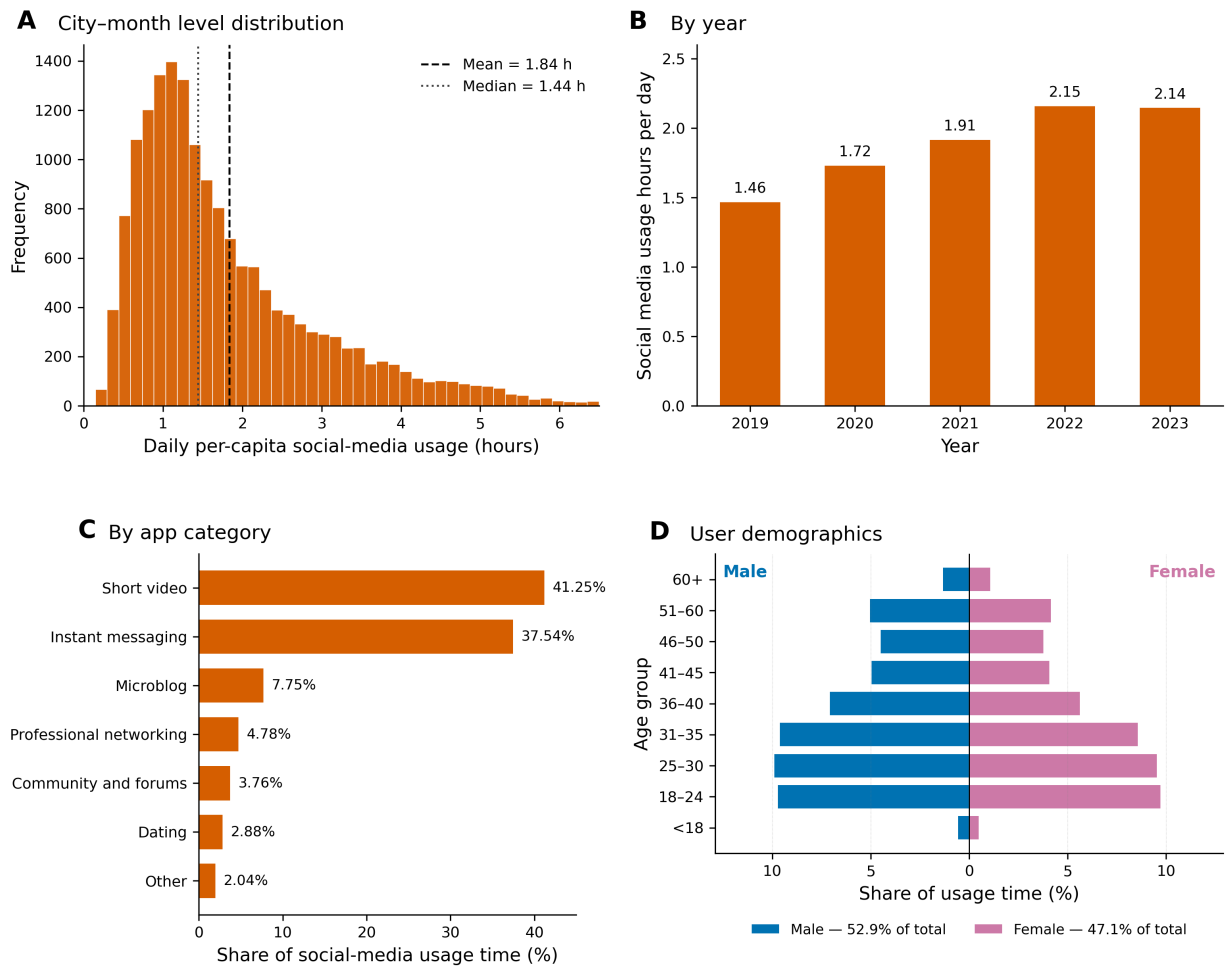


Figure S2: **Descriptive statistics of the population-scale digital-trace data on social media usage.** (**A**) City–month distribution of daily per-capita social media usage, measured in hours, based on 16,796 city–month observations across 323 prefecture-level cities from January 2019 through April 2023; dashed and dotted vertical lines mark the sample mean (1.84 hours) and median (1.44 hours). The distribution is right-skewed. (**B**) Annual mean daily per-capita social media usage. Mean daily usage increased from 1.46 hours in 2019 to 2.14 hours in January–April 2023. (**C**) Share of total usage time by app category. The short-video category accounts for the largest share of measured time (41.25%), followed by instant messaging (37.54%), microblog (7.75%), professional networking (4.78%), community and forum (3.76%), dating (2.88%), and other social-media apps (2.04%). (**D**) Population pyramid of usage time by gender (male, 52.9%; female, 47.1%) and age group; usage is concentrated among users aged 18–40 and is highest among those aged 18–30. Data are aggregated to city × month × gender × age cells by the data provider, as described in Section S1.2.

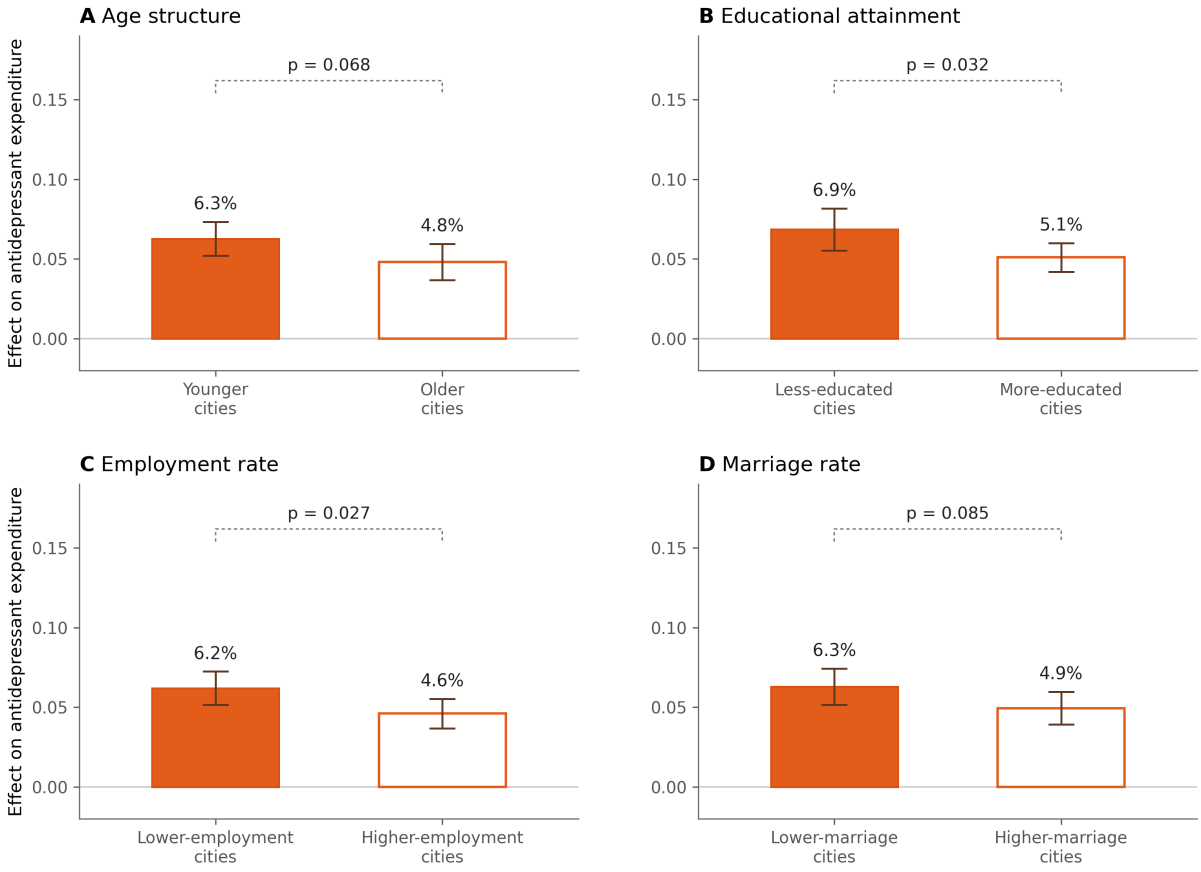


Figure S3: **OLS heterogeneity estimates by city characteristics.** OLS estimates of the association between daily per-capita social media usage and log per-capita antidepressant expenditure, splitting cities at the median of each 2015 characteristic. (**A**) Estimates by city age structure, comparing cities with above- versus below-median shares of residents aged 15–34. (**B**) Estimates by educational attainment, comparing cities with above- versus below-median shares of college-educated residents. (**C**) Estimates by employment rate, comparing cities with above- versus below-median employment rates among residents aged 16 and above. (**D**) Estimates by marriage rate, comparing cities with above- versus below-median marriage rates. Filled bars denote the more-vulnerable subsample, whereas hollow bars denote the comparison subsample. Error bars indicate 95% confidence intervals based on standard errors clustered at the city level. Brackets report p-values from tests of coefficient equality using a fully interacted OLS specification. All regressions include city and year-month fixed effects and time-varying city-level controls. Each split divides the full panel of 16,796 city–month observations into the two subsamples. The OLS estimates show a pattern similar to the IV estimates in Fig. 3.

Table S1: **Classification of antidepressant compounds as first-line or second-line.** This table lists the 17 generic compounds in the sample, together with their therapeutic class and assigned clinical line. First-line compounds are medications typically positioned as initial pharmacological options for new or mild-to-moderate depressive symptoms. Second-line compounds are medications more often positioned as follow-up options or reserved for complex, recurrent, or treatment-resistant cases. The classification is fixed at the compound level before estimation, does not vary across cities or time, and should be interpreted as a clinical treatment-positioning measure rather than a patient-level diagnosis or a direct measure of disease severity.

Compound	Therapeutic class
First-line (8 compounds)	
Fluoxetine	SSRI
Fluvoxamine	SSRI
Sertraline	SSRI
Paroxetine	SSRI
Escitalopram	SSRI
Citalopram	SSRI
Duloxetine	SNRI
Agomelatine	Melatonergic antidepressant
Second-line (9 compounds)	
Venlafaxine	SNRI
Milnacipran	SNRI
Mirtazapine	NaSSA
Mianserin	Tetracyclic
Reboxetine	NRI
Amitriptyline	TCA
Clomipramine	TCA
Doxepin	TCA
Flupentixol–Melitracen	TCA combination

Table S2: **Summary statistics: city–month sample.** This table reports summary statistics for the city–month panel used in the baseline and robustness analyses. The sample covers 323 prefecture-level cities in mainland China observed monthly from January 2019 through April 2023, yielding 16,796 city–month observations. The statistics are based on antidepressant procurement records, digital-trace data, and city-level socioeconomic information from the China City Statistical Yearbook. Population is measured in millions; GDP per capita and fiscal expenditure per capita are measured in thousands of RMB; hospital beds and doctors are measured per 1,000 residents. Per-capita antidepressant expenditure is measured in RMB per resident per month; all variables are reported in levels rather than logs.

Variable	Obs.	Mean	SD
Per-capita antidepressant expenditure, total	16,796	0.283	0.437
Per-capita antidepressant expenditure, first-line	16,796	0.191	0.339
Per-capita antidepressant expenditure, second-line	16,796	0.092	0.129
Per-capita social media usage time (hours/day)	16,796	1.836	1.262
Per-capita non-social-media app usage time (hours/day)	16,796	3.090	2.500
Population (million)	16,796	4.267	3.278
GDP per capita (thousand RMB)	16,796	74.317	61.497
Fiscal expenditure per capita (thousand RMB)	16,796	14.258	9.768
Share of secondary industry	16,796	0.393	0.101
Share of tertiary industry	16,796	0.491	0.083
Hospital beds per 1,000 residents	16,796	5.496	1.962
Doctors per 1,000 residents	16,796	3.098	1.291

Table S3: **Summary statistics: CGSS individual-level data.** This table reports summary statistics for the individual-level Chinese General Social Survey (CGSS) microdata used in the mechanism analysis. The sample pools the 2017, 2018, 2021, and 2023 waves. Panel A summarizes variables used in the analyses of time use and self-reported depression; Panel B summarizes variables used in the analyses of social perceptions. Statistics are calculated using the observations included in the corresponding regressions. The depression score and the time-use measures are coded on five-point scales; social media usage time is measured in daily per-capita hours in the respondent's corresponding city × gender × age cell; the construction of the social-perception measures is described in Section S1.6.

Variable	Obs.	Mean	SD
Panel A. Variables used in the time-use and self-reported depression analyses			
Depression score	34,311	2.279	1.166
Social media usage time	34,311	1.231	1.688
Internet use	34,311	3.072	1.895
Offline socializing	34,311	2.381	1.192
Rest	34,311	3.245	1.088
Offline exercising	34,311	2.666	1.618
Meeting friends	34,311	2.146	1.036
Offline shopping	34,311	2.599	1.145
Visiting relatives	34,311	2.067	0.732
Offline studying	34,311	1.910	1.094
Panel B. Variables used in the social-perception analyses			
Depression score	31,775	2.279	1.160
Social media usage time	31,775	1.251	1.708
Future pessimism	31,775	−0.287	1.661
Income underestimation	31,775	0.483	1.470
Perceived inequality	31,775	3.215	1.290
Social distrust	31,775	3.194	1.115

Table S4: **Predicted social media use strongly predicts observed social media use.** This table reports the first-stage regression of observed social media use on predicted social media use (equation (3) of the main text). Predicted social media use is constructed using the shift-share (Bartik) design described in Section S2.2. The specification includes city and year-month fixed effects and time-varying city-level controls. The first stage is common to all three outcome columns of Table 1, because neither the endogenous regressor nor the instrument varies with the outcome. The coefficient on predicted social media use is positive, large, and precisely estimated; the Kleibergen–Paap Wald F-statistic is well above conventional thresholds for weak instruments. Standard errors clustered at the city level are reported in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

	Social media usage (hours)
	(1)
Predicted social media use	0.8180***
	(0.0346)
City fixed effects	Yes
Year-month fixed effects	Yes
Controls	Yes
Kleibergen–Paap Wald F-statistic	559.9
Observations	16,796

Table S5: **Reduced-form estimates of predicted social media use on antidepressant expenditure.** This table reports reduced-form estimates of predicted social media use on log per-capita antidepressant expenditure. Predicted social media use is constructed using the shift-share (Bartik) design described in Section S2.2. Columns 1 to 3 report results for total expenditure, first-line expenditure, and second-line expenditure, respectively. All specifications include city and year-month fixed effects and time-varying city-level controls. Standard errors clustered at the city level are reported in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

	Log per-capita antidepressant expenditure		
	Total	First-line	Second-line
	(1)	(2)	(3)
Predicted social media use	0.0634***	0.0638***	0.0049**
	(0.0050)	(0.0047)	(0.0022)
City fixed effects	Yes	Yes	Yes
Year-month fixed effects	Yes	Yes	Yes
Controls	Yes	Yes	Yes
Observations	16,796	16,796	16,796

Table S6: **Drug-by-drug OLS estimates of the effect of social media use on per-capita antidepressant dosage.** This table reports separate OLS regressions for each generic compound. The dependent variable is the natural logarithm of per-capita dosage in milligrams. Panel A reports first-line compounds; Panel B reports second-line compounds. These estimates are displayed graphically in Fig. 1A. All specifications include city and year-month fixed effects and time-varying city-level controls. Standard errors clustered at the city level are reported in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Panel A. First-line drugs

	Sertraline	Fluvoxamine	Paroxetine	Fluoxetine	Escitalopram	Citalopram	Agomelatine	Duloxetine
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Social media usage time	0.102***	0.071***	0.062***	0.045***	0.035***	0.034***	0.036***	0.034***
	(0.008)	(0.007)	(0.005)	(0.005)	(0.003)	(0.003)	(0.003)	(0.003)
City fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year-month fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	16,796	16,796	16,796	16,796	16,796	16,796	16,796	16,796

Panel B. Second-line drugs

	Venlafaxine	Mirtazapine	Flupentixol–Melitracen	Clomipramine	Doxepin	Mianserin	Milnacipran	Reboxetine	Amitriptyline
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Social media usage time	0.028***	0.011***	0.001	0.001	0.000	0.001***	0.001**	−0.000	−0.000
	(0.006)	(0.004)	(0.003)	(0.001)	(0.001)	(0.001)	(0.001)	(0.000)	(0.001)
City fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year-month fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	16,796	16,796	16,796	16,796	16,796	16,796	16,796	16,796	16,796

Table S7: **Drug-by-drug 2SLS estimates of the effect of social media use on per-capita antidepressant dosage.** This table reports separate 2SLS regressions for each generic compound, using predicted social media use as an instrument for observed social media use. Predicted social media use is constructed using the shift-share (Bartik) design described in Section S2.2. The dependent variable is the natural logarithm of per-capita dosage in milligrams. Panel A reports first-line compounds; Panel B reports second-line compounds. These estimates are displayed graphically in Fig. 1B. All specifications include city and year-month fixed effects and time-varying city-level controls. Standard errors clustered at the city level are reported in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Panel A. First-line drugs

	Sertraline	Fluvoxamine	Paroxetine	Fluoxetine	Escitalopram	Citalopram	Agomelatine	Duloxetine
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Social media usage time	0.157***	0.111***	0.088***	0.065***	0.055***	0.051***	0.049***	0.049***
	(0.011)	(0.010)	(0.007)	(0.007)	(0.005)	(0.005)	(0.006)	(0.004)
City fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year-month fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Kleibergen–Paap Wald F-statistic	559.9	559.9	559.9	559.9	559.9	559.9	559.9	559.9
Observations	16,796	16,796	16,796	16,796	16,796	16,796	16,796	16,796

Panel B. Second-line drugs

	Venlafaxine	Mirtazapine	Flupentixol–Melitracen	Clomipramine	Doxepin	Mianserin	Milnacipran	Reboxetine	Amitriptyline
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Social media usage time	0.046***	0.014**	0.008	0.004*	0.001	0.001	0.000	−0.000	−0.000
	(0.012)	(0.006)	(0.005)	(0.002)	(0.002)	(0.001)	(0.001)	(0.000)	(0.002)
City fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year-month fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Kleibergen–Paap Wald F-statistic	559.9	559.9	559.9	559.9	559.9	559.9	559.9	559.9	559.9
Observations	16,796	16,796	16,796	16,796	16,796	16,796	16,796	16,796	16,796

Table S8: **Placebo test using non-social-media app use.** This table reports OLS and 2SLS estimates of the effect of daily per-capita non-social-media app usage (hours) on log per-capita antidepressant expenditure. Column 1 reports OLS estimates. Column 2 reports 2SLS estimates, instrumenting non-social-media app usage with its corresponding shift-share instrument. Both specifications include city and year-month fixed effects and time-varying city-level controls. Standard errors clustered at the city level are reported in parentheses. The Kleibergen–Paap Wald F-statistic is reported for the 2SLS specification. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

	Log per-capita antidepressant expenditure	
	OLS	2SLS
	(1)	(2)
Non-social-media app usage time	0.0011	0.0013
	(0.0028)	(0.0043)
City fixed effects	Yes	Yes
Year-month fixed effects	Yes	Yes
Controls	Yes	Yes
Kleibergen–Paap Wald F-statistic	/	111.4
Observations	16,796	16,796

Table S9: **Placebo test using later-period predicted social media use.** This table reports placebo regressions of log per-capita antidepressant expenditure observed earlier in the sample on predicted social media use constructed for 2022. Columns 1, 2, and 3 use antidepressant expenditure in 2019, 2020, and 2021, respectively. The regressor is the Bartik instrument defined in equation (S2), with each outcome month matched to predicted social media use in the corresponding calendar month of 2022. The estimates are small and statistically insignificant across all three years, indicating that later predicted social media use does not systematically predict earlier antidepressant expenditure. All specifications include city and year-month fixed effects and time-varying city-level controls. Standard errors clustered at the city level are reported in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

	Log per-capita antidepressant expenditure		
	2019	2020	2021
	(1)	(2)	(3)
Later-period predicted social media use	0.0085	−0.0020	0.0060
	(0.0055)	(0.0046)	(0.0071)
City fixed effects	Yes	Yes	Yes
Year-month fixed effects	Yes	Yes	Yes
Controls	Yes	Yes	Yes
Observations	3,876	3,876	3,876

Table S10: **Robustness to differential trends in predetermined city characteristics.** This table reports 2SLS estimates of the effect of social media usage time on log per-capita antidepressant expenditure, augmenting the baseline specification with interactions between year-month fixed effects and predetermined city-level characteristics measured in 2015. Column 1 interacts year-month fixed effects with the 2015 gender × age population shares that construct the instrument. Columns 2 to 7 interact year-month fixed effects with the share of college-educated residents, the employment rate among residents aged 16 and above, the share of married residents, population, GDP per capita, and mobile penetration, respectively. Column 8 includes all seven sets of interactions jointly. The estimates remain stable in magnitude and statistical significance across columns, indicating that the baseline results are not driven by differential trends associated with predetermined demographic composition, socioeconomic conditions, or technology penetration. All specifications include city and year-month fixed effects and time-varying city-level controls. Standard errors clustered at the city level are reported in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

	Log per-capita antidepressant expenditure							
	All gender × age shares	College	Employed	Married	Population	GDP per capita	Mobile penetration	All jointly
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Social media usage time	0.0833***	0.0775***	0.0772***	0.0780***	0.0775***	0.0754***	0.0764***	0.0830***
	(0.0061)	(0.0053)	(0.0054)	(0.0053)	(0.0053)	(0.0051)	(0.0054)	(0.0061)
City fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year-month fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Characteristic × Year-Month	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Kleibergen–Paap Wald F-statistic	431.6	568.2	555.8	575.8	560.9	550.8	540.7	405.1
Observations	16,796	16,796	16,796	16,796	16,796	16,796	16,796	16,796

Table S11: **Balance of predetermined city characteristics and predicted social media use.** This table reports city-level balance tests relating predetermined 2015 city characteristics to each city's average predicted social media use over the January 2019–April 2023 sample period. Columns 1 and 2 separately regress average predicted social media use on GDP per capita and fiscal expenditure per capita in 2015. Column 3 includes 2015 secondary and tertiary industry shares. Column 4 includes 2015 medical-resource variables, measured per 1,000 residents. Column 5 includes all six predetermined city characteristics jointly. Standard errors clustered at the city level are reported in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

	Predicted social media use				
	(1)	(2)	(3)	(4)	(5)
GDP per capita, 2015	0.0009				0.0003
	(0.0006)				(0.0008)
Fiscal expenditure per capita, 2015		0.0019			−0.0026
		(0.0033)			(0.0031)
Share of secondary industry, 2015			−0.0316		−0.4974
			(0.4753)		(0.5112)
Share of tertiary industry, 2015			0.2696		−0.6641
			(0.4376)		(0.5778)
Hospital beds per 1,000 residents, 2015				0.0167	0.0258
				(0.0294)	(0.0306)
Doctors per 1,000 residents, 2015				0.0450	0.0550
				(0.0365)	(0.0376)
Observations	323	323	323	323	323

Table S12: **Rotemberg weights for the components of predicted social media use.** This table reports the five largest Rotemberg weights for the gender × age components underlying predicted social media use. Each component is the interaction between a city's 2015 population share in a gender × age cell and the leave-one-out social media usage of that same cell in other cities of the same province. Weights are computed using the baseline IV specification after residualizing the endogenous variable, outcome, and component instruments with respect to city fixed effects, year-month fixed effects, and city-level controls. Cumulative weight reports the running sum of the listed weights. The just-identified IV estimate reports the IV coefficient obtained using each component instrument separately. Across all 18 components, all Rotemberg weights are positive, and the largest weight is 9.5%. The five largest weights sum to 45.7%, while the component-specific just-identified IV estimates across all 18 components range from 0.068 to 0.082.

Gender × age cell	Rotemberg weight	Cumulative weight	Just-identified IV estimate
Male, 31–35	0.095	0.095	0.082
Male, 25–30	0.094	0.189	0.081
Male, 18–24	0.090	0.279	0.080
Female, 25–30	0.090	0.369	0.082
Female, 18–24	0.088	0.457	0.082

Table S13: **Robustness to an alternative instrument.** This table reports 2SLS estimates of the effect of social media usage time on log per-capita antidepressant expenditure, using an alternative Bartik instrument constructed from usage among the same demographic group in other cities nationwide rather than in other cities of the same province, excluding the focal city's own contribution when constructing the alternative Bartik instrument. Columns 1 to 3 report results for total, first-line, and second-line expenditure, respectively. The estimates are quantitatively similar to the baseline estimates reported in Table 1, with a Kleibergen–Paap Wald F-statistic of 20.8. All specifications include city and year-month fixed effects and time-varying city-level controls. Standard errors clustered at the city level are reported in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

	Log per-capita antidepressant expenditure		
	Total	First-line	Second-line
	(1)	(2)	(3)
Social media usage time	0.0740***	0.0725***	0.0080
	(0.0164)	(0.0151)	(0.0082)
City fixed effects	Yes	Yes	Yes
Year-month fixed effects	Yes	Yes	Yes
Controls	Yes	Yes	Yes
Kleibergen–Paap Wald F-statistic	20.8	20.8	20.8
Observations	16,796	16,796	16,796

Table S14: **Additional robustness checks.** This table reports 2SLS estimates of the effect of social media usage time on log per-capita antidepressant expenditure under a series of additional robustness checks. Column 1 weights observations by city population. Column 2 excludes calendar months containing major public holidays, during which both social media usage and healthcare procurement may be atypical. Column 3 excludes the four municipalities of Beijing, Shanghai, Tianjin, and Chongqing. Column 4 clusters standard errors at the province level. Column 5 augments the baseline specification with city-specific linear time trends. Column 6 includes city–year fixed effects. Across all six specifications, the estimated effect remains positive and statistically significant. All specifications include or absorb city fixed effects, year-month fixed effects, and controls. Standard errors are clustered at the city level except in column 4, where they are clustered at the province level, and are reported in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

	Log per-capita antidepressant expenditure					
	Pop. weighted	Excl. holidays	Excl. municipalities	Province clustering	City linear trends	City–year FE
	(1)	(2)	(3)	(4)	(5)	(6)
Social media usage time	0.0953***	0.0828***	0.0770***	0.0775***	0.0908***	0.0498***
	(0.0067)	(0.0056)	(0.0054)	(0.0121)	(0.0067)	(0.0054)
City fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Year-month fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes	Yes	Yes
City linear trends	No	No	No	No	Yes	No
City–year fixed effects	No	No	No	No	No	Yes
Kleibergen–Paap Wald F-statistic	523.2	484.3	555.7	376.7	324.0	326.3
Observations	16,796	10,982	16,588	16,796	16,796	16,796

Table S15: **Robustness to the COVID-19 pandemic.** This table reports 2SLS estimates of the effect of social media usage time on log per-capita antidepressant expenditure, addressing the concern that the COVID-19 pandemic may have affected both social media use and antidepressant expenditure. Column 1 controls for an indicator for city–months under citywide lockdown, following the existing literature (*2*). Column 2 excludes the COVID-period years (2020–2022). The coefficient on social media usage time remains stable in both specifications, and the lockdown indicator enters negatively, consistent with restricted access to medical institutions during lockdowns. All specifications include city and year-month fixed effects and time-varying city-level controls. Standard errors clustered at the city level are reported in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

	Log per-capita antidepressant expenditure	
	Controlling for citywide lockdowns	Excluding the COVID period
	(1)	(2)
Social media usage time	0.0774***	0.0748***
	(0.0053)	(0.0079)
Citywide lockdown	–0.0284**	
	(0.0118)	
City fixed effects	Yes	Yes
Year-month fixed effects	Yes	Yes
Controls	Yes	Yes
Kleibergen–Paap Wald F-statistic	559.3	405.5
Observations	16,796	5,168

Table S16: **Robustness to public perceptions of unemployment and recession.** This table examines whether public perceptions of unemployment and recession threaten the IV identification strategy. The unemployment perception and recession perception indices are city–month measures constructed as the first principal component of the corresponding keyword-level Baidu search volumes and standardized as z-scores with mean zero and unit standard deviation over the city–month sample. Columns 1 and 2 report 2SLS estimates of the effect of social media usage on log per-capita antidepressant expenditure while controlling separately for the unemployment perception and recession perception indices. Columns 3 and 4 regress the unemployment perception index and recession perception index, respectively, on predicted social media use. Predicted social media use is constructed using the shift-share (Bartik) design described in Section S2.2. All specifications include city and year-month fixed effects and time-varying city-level controls. Standard errors clustered at the city level are reported in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

	Log per-capita antidepressant expenditure		Unemployment perception index	Recession perception index
	(1)	(2)	(3)	(4)
Social media usage time	0.0775***	0.0775***		
	(0.0053)	(0.0053)		
Unemployment perception index	0.0074			
	(0.0130)			
Recession perception index		0.0203		
		(0.0207)		
Predicted social media use			0.0001	−0.0001
			(0.0112)	(0.0044)
Estimation	2SLS	2SLS	OLS	OLS
City fixed effects	Yes	Yes	Yes	Yes
Year-month fixed effects	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes
Kleibergen–Paap Wald F-statistic	595.9	574.0	/	/
Observations	16,796	16,796	16,796	16,796

Table S17: **Mechanism estimates: time displacement.** This table reports descriptive OLS estimates using individual-level data from the 2017, 2018, 2021, and 2023 CGSS waves. Panel A regresses self-reported depression on each of the eight time-use measures; Panel B regresses each measure on daily per-capita social media use in the respondent's city-by-gender-by-age cell. These estimates are displayed graphically in Fig. 4, A and B. Self-reported depression and the time-use measures are coded on five-point frequency scales, with higher values indicating greater frequency. All specifications include controls for gender, age, rural *hukou* status, minority ethnicity, education, and employment; city and year fixed effects; and time-varying city-level controls. Standard errors clustered at the city level are reported in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Panel A. Association between time use and self-reported depression

	Internet use	Offline socializing	Rest	Offline exercising	Meeting friends	Offline shopping	Visiting relatives	Offline studying
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Time-use measure	0.082***	−0.123***	−0.124***	−0.056***	−0.101***	−0.081***	−0.105***	−0.079***
	(0.007)	(0.010)	(0.012)	(0.005)	(0.010)	(0.006)	(0.011)	(0.008)
Individual controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
City fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
City-level controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	34,311	34,311	34,311	34,311	34,311	34,311	34,311	34,311

Panel B. Association between social media use and time-use measures

	Internet use	Offline socializing	Rest	Offline exercising	Meeting friends	Offline shopping	Visiting relatives	Offline studying
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Social media usage time	0.207***	−0.115***	−0.097***	−0.073***	−0.058***	−0.050***	−0.026***	−0.016**
	(0.024)	(0.012)	(0.010)	(0.012)	(0.006)	(0.009)	(0.005)	(0.008)
Individual controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
City fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
City-level controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	34,311	34,311	34,311	34,311	34,311	34,311	34,311	34,311

Table S18: **Mechanism estimates: social perceptions.** This table reports descriptive OLS estimates using individual-level data from the 2017, 2018, 2021, and 2023 CGSS waves. Panel A regresses self-reported depression on each of the four social perception measures; Panel B regresses each measure on daily per-capita social media use in the respondent's city-by-gender-by-age cell. These estimates are displayed graphically in Fig. 4, C and D. Higher values indicate greater future pessimism, income underestimation, perceived inequality, and social distrust. All specifications include controls for gender, age, rural *hukou* status, minority ethnicity, education, and employment; city and year fixed effects; and time-varying city-level controls. Standard errors clustered at the city level are reported in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Panel A. Association between social perceptions and self-reported depression

	Future pessimism	Income underestimation	Perceived inequality	Social distrust
	(1)	(2)	(3)	(4)
Social-perception measure	0.060***	0.041***	0.184***	0.122***
	(0.008)	(0.006)	(0.012)	(0.008)
Individual controls	Yes	Yes	Yes	Yes
City fixed effects	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes
City-level controls	Yes	Yes	Yes	Yes
Observations	31,775	31,775	31,775	31,775

Panel B. Association between social media use and social perceptions

	Future pessimism	Income underestimation	Perceived inequality	Social distrust
	(1)	(2)	(3)	(4)
Social media usage time	0.115***	0.112***	0.104***	0.032***
	(0.013)	(0.014)	(0.013)	(0.011)
Individual controls	Yes	Yes	Yes	Yes
City fixed effects	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes
City-level controls	Yes	Yes	Yes	Yes
Observations	31,775	31,775	31,775	31,775

Table S19: **Depression awareness and antidepressant expenditure.** This table reports city–month estimates using the depression-related Baidu search index as a proxy for depression awareness. The index is constructed as the first principal component of the keyword-level Baidu search volumes and standardized as a z-score with mean zero and unit standard deviation over the city–month sample. Column 1 estimates the association between the Baidu search index and the natural logarithm of per-capita antidepressant expenditure. Column 2 adds observed social media usage to the OLS specification. Column 3 reports the corresponding 2SLS specification, using predicted social media use as an instrument for observed social media use. Predicted social media use is constructed using the shift-share (Bartik) design described in Section S2.2. All specifications include city and year-month fixed effects and time-varying city-level controls. Standard errors clustered at the city level are reported in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

	Log per-capita antidepressant expenditure		
	OLS	OLS	2SLS
	(1)	(2)	(3)
Social media usage time		0.0547***	0.0765***
		(0.0037)	(0.0052)
Depression-related Baidu search index (z-score)	0.0942***	0.0666***	0.0556***
	(0.0175)	(0.0139)	(0.0129)
City fixed effects	Yes	Yes	Yes
Year-month fixed effects	Yes	Yes	Yes
Controls	Yes	Yes	Yes
Kleibergen–Paap Wald F-statistic	/	/	563.2
Observations	16,796	16,796	16,796

Table S20: **Social media usage and self-reported depression.** This table reports OLS estimates using individual-level data from the 2017, 2018, 2021, and 2023 CGSS waves. The dependent variable is self-reported depression. Daily per-capita social media use is measured for the respondent's corresponding city-by-gender-by-age cell. The specification includes controls for gender, age, rural *hukou* status, minority ethnicity, education, and employment; city and year fixed effects; and time-varying city-level controls. Standard errors clustered at the city level are reported in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

	Self-reported depression
	(1)
Social media usage time	0.101***
	(0.015)
Individual controls	Yes
City fixed effects	Yes
Year fixed effects	Yes
City-level controls	Yes
Observations	34,311

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